

Information with SVGD

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1 Direct Gradient through Uniform Sampling

Where S is shorthand for *softmax*. Directly approximate the integral through uniform samples $x_t \sim U(\mathcal{X})$.

$$\nabla_{\omega} \log p_{\theta}(m_t|x_T) = \nabla_{\omega} \log \int_{x_t \in \mathcal{X}} p_{\theta}(m_t|x_t) p_{\theta}(x_t|x_T) \quad (1)$$

$$\stackrel{s}{\approx} \nabla_{\omega} \log \text{const} \sum_{x_t \sim U(\mathcal{X})} \mathbb{P}(m_t|x_t) \mathbb{P}(x_t|x_T) \quad (2)$$

$$= \nabla_{\omega} \log \sum_{x_t \sim U(\mathcal{X})} p_{\theta}(m_t|x_t) \frac{\exp(f_{\theta}(x_t, x_T))}{\int_{\hat{x}_t \in \mathcal{X}} \exp(f_{\theta}(\hat{x}_t, x_T))} \quad (3)$$

$$\stackrel{s}{\approx} \nabla_{\omega} \log \sum_{x_t \sim U(\mathcal{X})} p_{\theta}(m_t|x_t) \frac{\exp(f_{\theta}(x_t, x_T))}{\sum_{\hat{x}_t \sim U(\mathcal{X})} \exp(f_{\theta}(\hat{x}_t, x_T))} \quad (4)$$

$$= \nabla_{\omega} \log \left(\frac{\sum_{x_t \sim U(\mathcal{X})} p_{\theta}(m_t|x_t) \exp(f_{\theta}(x_t, x_T))}{\sum_{\hat{x}_t \sim U(\mathcal{X})} \exp(f_{\theta}(\hat{x}_t, x_T))} \right) \quad (5)$$

$$= \nabla_{\omega} \left[\log \sum_{x_t \sim U(\mathcal{X})} p_{\theta}(m_t|x_t) e^{f_{\theta}(x_t, x_T)} - \log \sum_{x_t \sim U(\mathcal{X})} e^{f_{\theta}(x_t, x_T)} \right] \quad (6)$$

$$= S_{x_t \sim U} [\log p_{\theta}(m_t|x_t) + f_{\theta}(x_t, x_T)] \cdot \nabla_{\omega} (\log p_{\theta}(m_t|x_t) + f_{\theta}(x_t, x_T)) \quad (7)$$

$$- S_{x_t \sim U} [f_{\theta}(x_t, x_T)] \cdot \nabla_{\omega} (f_{\theta}(x_t, x_T)) \quad (8)$$

2 SGVD Sampling

Continue from Uniform Sampling:

$$7 = S_{x_t \sim U} [\log p_\theta(m_t|x_t) + f_\theta(x_t, x_T)] \cdot \nabla_\omega (\log p_\theta(m_t|x_t) + f_\theta(x_t, x_T)) \quad (9)$$

$$= \frac{1}{\alpha} S_{x_t \sim U} [\alpha (\log p_\theta(m_t|x_t) + f_\theta(x_t, x_T)) + (1-\alpha) (\log p_{\hat{\theta}}(m_t|x_t) + f_{\hat{\theta}}(x_t, \hat{x}_T))] \quad (10)$$

$$\cdot \nabla_\omega (\alpha (\log p_\theta(m_t|x_t) + f_\theta(x_t, x_T))) \quad (11)$$

$$= \frac{1}{\alpha} \nabla_\omega \left(\log \sum_{x_t \sim U} e^{\alpha (\log p_\theta(m_t|x_t) + f_\theta(x_t, x_T)) + (1-\alpha) (\log p_{\hat{\theta}}(m_t|x_t) + f_{\hat{\theta}}(x_t, \hat{x}_T))} \right) \quad (12)$$

$$= \frac{1}{\alpha} \nabla_\omega \left(\log \sum_{x_t \sim U} e^{\alpha (\log p_\theta(m_t|x_t) + f_\theta(x_t, x_T)) + (1-\alpha) \log p_{\hat{\theta}}(m_t|x_t)} \frac{e^{(1-\alpha) f_{\hat{\theta}}(x_t, \hat{x}_T)}}{\sum_{\hat{x}_t \sim U} e^{(1-\alpha) f_{\hat{\theta}}(\hat{x}_t, \hat{x}_T)}} \right) \quad (13)$$

$$\stackrel{s}{\approx} \frac{1}{\alpha} \nabla_\omega \left(\log \sum_{x_t \sim MC_{(1-\alpha)}(\hat{x}_T)} e^{\alpha (\log p_\theta(m_t|x_t) + f_\theta(x_t, x_T)) + (1-\alpha) \log p_{\hat{\theta}}(m_t|x_t)} \right) \quad (14)$$

$$= S_{x_t \sim MC_{(1-\alpha)}(\hat{x}_T)} [\log p_\theta(m_t|x_t) + \alpha f_\theta(x_t, x_T)] \cdot \nabla_\omega (\log p_\theta(m_t|x_t) + f_\theta(x_t, x_T)) \quad (15)$$

$$\lim_{\alpha \rightarrow 0} = S_{x_t \sim MC} [\log p_\theta(m_t|x_t)] \cdot \nabla_\omega (\log p_\theta(m_t|x_t) + f_\theta(x_t, x_T)) \quad (16)$$

And the normalization term by the same logic:

$$8 = S_{x_t \sim U} [f_\theta(x_t, x_T)] \cdot \nabla_\omega (f_\theta(x_t, x_T)) \quad (17)$$

$$= S_{x_t \sim MC_{(1-\alpha)}(\hat{x}_T)} [\alpha f_\theta(x_t, x_T)] \cdot \nabla_\omega (f_\theta(x_t, x_T)) \quad (18)$$

$$\lim_{\alpha \rightarrow 0} = S_{x_t \sim MC} [0] \cdot \nabla_\omega (f_\theta(x_t, x_T)) \quad (19)$$

$$= \frac{1}{N_i} \sum_i^{N_i} \nabla_\omega (f_\theta(x_{t,i}, x_T)) \quad (20)$$

So the final gradient can be written as (16)-(20)